

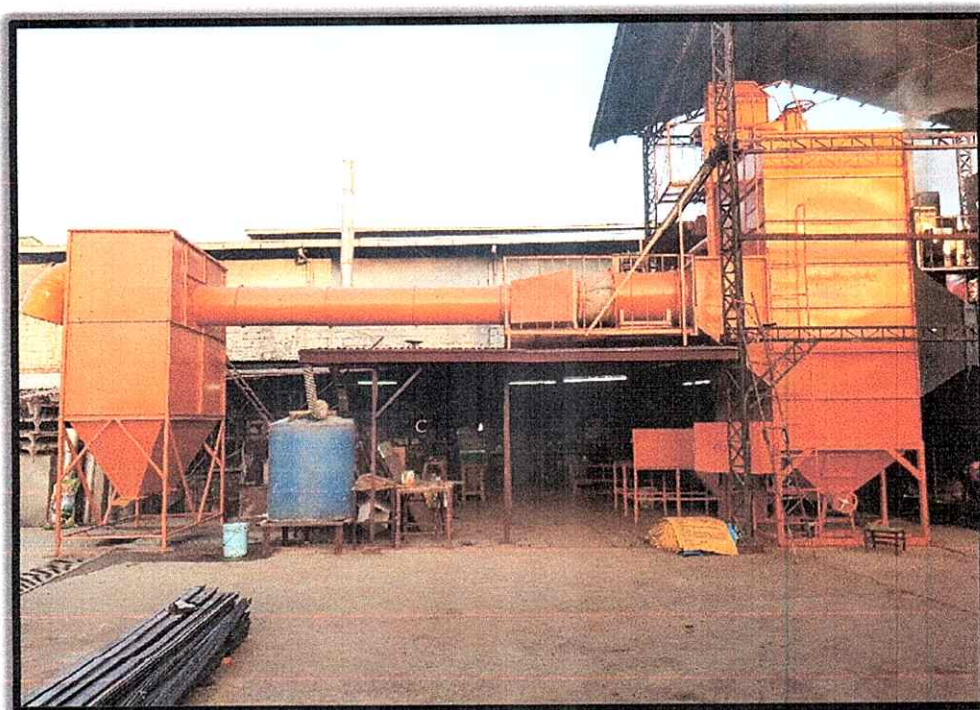


AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER



Test Report No. 2021 – 0593



Agricultural Machinery – Heated-Air Mechanical Grain Dryer LAKAS KULIGLIG RMD-LK66

Test Report Number	2021 – 0593
Date of Issuance	December 1, 2021
Valid Until	December 1, 2026
Agricultural Machinery	Heated-Air Mechanical Grain Dryer
Type	Batch, Recirculating, Dual heating system (Indirect-fired biomass furnace and direct-fired diesel-fed burner), Tested for rice using the biomass furnace
Brand	LAKAS KULIGLIG
Model	RMD-LK66
Test Requested by	P.I. FARM PRODUCTS, INC. 35F Alarcon St. Sitio Malabo, Maysan, Valenzuela City

SUMMARY

Performance Criteria	Requirement as per PNS/PAES 201:2015	AMTEC Test
Moisture Gradient, %, maximum	4.0 (batch type)	1.20
Drying Rate, %/h	0.8 (batch type)	2.18
Product Quality ^a		
a) Cracked Grain, % increase, maximum	3.0	3.75
b) Head rice, % decrease, maximum	5.0	2.51
c) Hulled/Damaged Grain, % increase, maximum	2.0	0
Drying Loss/Spillage, %, maximum	1.0	0
Drying Efficiency, %, minimum	75	85.43
Heating System Efficiency, %, minimum	50 ^b	35.74
Blower flow rate, cmm/kg (cmf/ton)	0.05 (1750)	0.028 (998.41)
Static Pressure, mm H ₂ O (in H ₂ O)	25 (1)	26.05 (1.03)
Grain Holding Capacity, kg	6000 ^c	5905.4
Discoloration, Traces of Unburned Fuel, Ashes on Grain Surface	Shall not be present	None
Fermented or Musty Smell	Shall not be present	None
Thermometer	Shall be provided	Provided
Pressure Gauge	Shall be provided	Not provided
Noise Level, dB(A), maximum	92	78.2

^a Allowable difference between the laboratory analysis and machine's performance

^b Indirect-fired biomass-fuel.

^c As per manufacturer's specification

OBSERVATIONS

1. Operator's manual	Not provided
2. Previous test reports of similar brand and model	2018-0457, 2019-0169, 2020-0589, 2021-0106
3. Manufacturer	P.I. FARM PRODUCTS, INC.
4. Heat source	Biomass Furnace
5. Safety controls/devices	The dryer has an automatic temperature controller with drying air temperature display.
6. Instrumentation, setting and adjustments	The dryer has a main control panel which contains all the push buttons, toggle switches, and digital controllers.
7. Loading and unloading	The dryer is provided with loading hopper, bucket elevator, and screw conveyor for loading and unloading of grains. It is also equipped with a bin for bagging.
8. Number of operators needed to load the grains	Three (one loader and two haulers)
9. Number of operators needed to load the biomass (rice hull)	One
10. Shall have adequate provision for fire control	None
11. Shall have adequate protection from or for all moving parts	Some moving parts have covers.
12. Shall be provided with features for access to parts during repair, maintenance, and operation	Parts are accessible for repair, maintenance, and operation.
13. Provision shall be made for dust control. The dust emission of the machine shall be within the acceptable limits set by the Department of Environment and Natural Resources.	Dust collection system is provided which consists of suction fans and cyclones.
14. Shall be free from defects that may be detrimental to its use	The dryer is free from manufacturing defects.
15. Shall be free from sharp edges and surfaces that may hurt the operator	The dryer is free from sharp edges and surfaces.
16. All metal parts should be machine bent, pressed and cut to avoid rough surfaces. All rough surfaces should be machine finished and smoothed.	All parts are machine fabricated.
17. The warning notice shall be in accordance with PNS/PAES 101:2000 – Agricultural Machinery – Technical Means for Ensuring Safety – General.	No warning notices were placed on the machine.

OBSERVATIONS (Cont'n)

18. Uniformity of parts and components for same brand and model must be maintained.	Not observed
19. Provision shall be made so that it can withstand extreme weather conditions to which it can be exposed.	The drying bin is made of painted steel materials.
20. Other remarks	a. The dryer has a toggle switch control on the control panel to use either the biomass furnace or diesel burner to heat the dryer. b. The biomass furnace was used as the primary heating system during the testing operation. c. The dryer was tested with only 5905.4 kg input test material, but there is still space left in the drying chamber as shown in Figure 4.

PURPOSE AND SCOPE OF TEST

Purpose	Performance Testing / Commercial
Date of Test	November 18, 2021
Location of Test	P.I. FARM PRODUCTS, INC. Manufacturing Plant 35F. Alarcon Street, Sitio Malabo, Maysan, Valenzuela City 14°42'10.4" N, 120°58'23.2" E
Items Tested	Drying Capacity, Moisture Reduction Rate, Heating System Efficiency, Drying Efficiency, Drying System Efficiency, Moisture Content Variation, Moisture Content Gradient, Fuel Consumption, Power Consumption, Number of Operators, and Quality of Dried Grains

METHODS OF TEST

The corn dryer was tested based on the PNS/PAES 202:2015 Agricultural Machinery – Heated-air Mechanical Grain Dryer – Methods of Test.

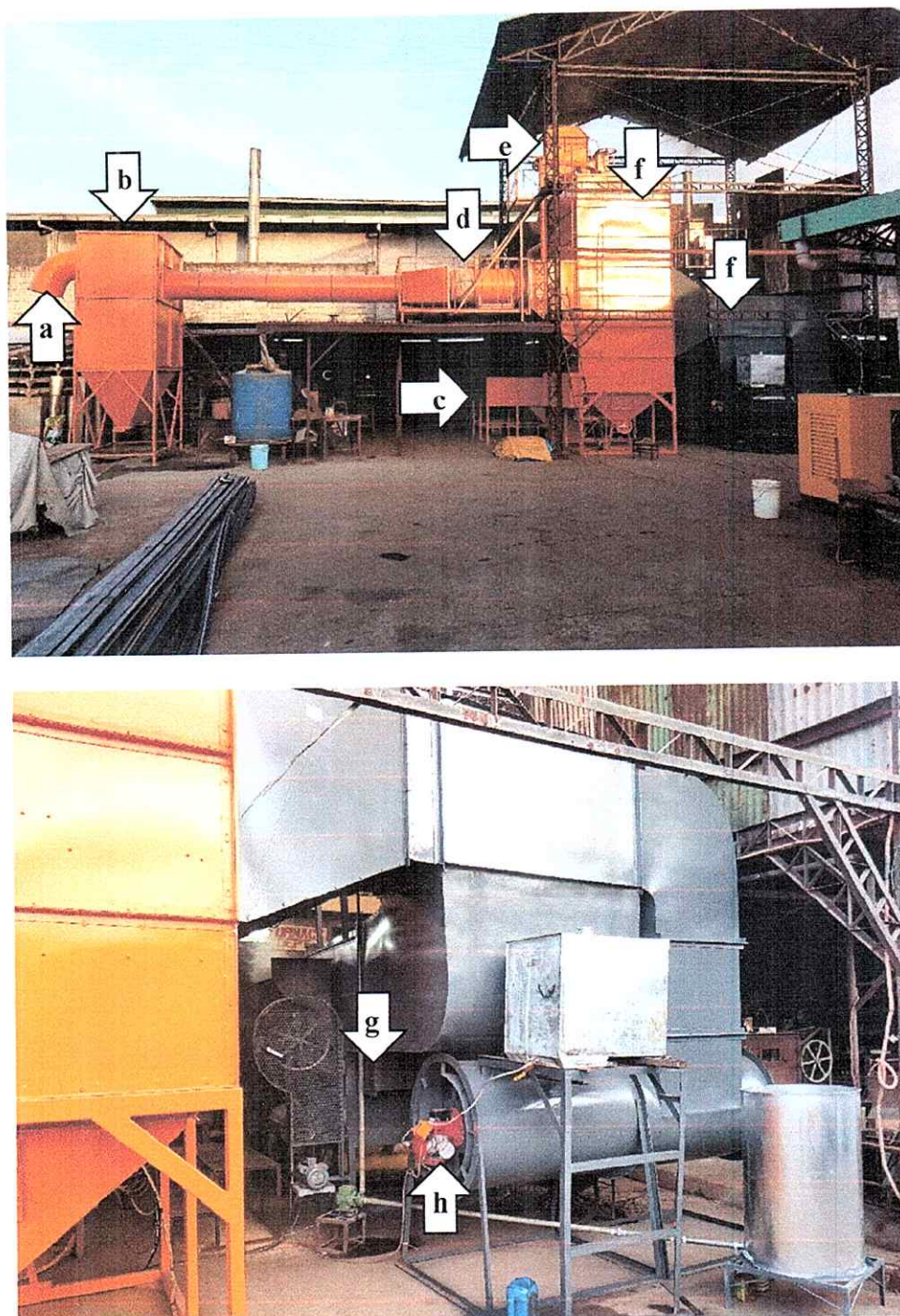
DESCRIPTION OF THE MACHINE

Figure 1. Different parts of the LAKAS KULIGLIG RMD-LK66 Heated-Air Mechanical Grain Dryer powered by Biomass Furnace include the following: (a) Exhaust air outle, (b) Cyclone/Dust collection system, (c) Loading hopper/table, (d) Fan, (e) Elevator, (f) Drying chamber, (f) Heated-air duct, (g) Biomass furnace, and (h) Diesel burner.

DESCRIPTION OF THE MACHINE (Cont'n)



Figure 2. Different parts of the LAKAS KULIGLIG RMD-LK66 Heated-Air Mechanical Grain Dryer powered by Biomass Furnace include the following:
(a) Bagging bin, (b) Control panel,
and (c) Loading hopper/table.

SPECIFICATIONS

Item		Manufacturer's Specification ^a	AMTEC Verification
1	Drying rate, %/h	ND	2.18 ^b
2	Overall dimensions, mm (installed)		
2.1	Length	13980	17530
2.2	Width	4340	4340
2.3	Height	7500	7500
3	Drying bin		
3.1	Type	Vertical, Rectangular, LSU	Vertical, Rectangular, LSU
3.2	Dimensions, mm		
3.2.1	Length	2830	2830
3.2.2	Width/Diameter	1830	1830
3.2.3	Height	21946	21946
3.3	Holding capacity, kg	6000	5905.4 ^b
3.4	Maximum grain depth, mm	ND	NM
3.5	Material		
3.5.1	Frame	Angle bar, 2 mm G.I. sheet	Angle bar
3.5.2	Holding bin	Angle bar, 2 mm G.I. sheet	Angle bar, Steel sheet
3.5.3	Wall	Angle bar, 2 mm G.I. sheet	Angle bar, Steel sheet
4	Fan		
4.1	Type	In-line centrifugal	In-line centrifugal
4.2	Brand and model	SIEMENS YL 90 S-4	SIEMENS YL 90 S-4
4.3	Air flow rate, m ³ /min	ND	166.96 ^b
4.4	Static pressure, mmH ₂ O	25.0	26.05 ^b
4.5	Material	B.I. sheet, G.I. sheet	Steel sheet
4.6	Number	2	2
4.6	Prime mover	3.73 kW, 220V, TECO single phase electric motor	3.73 kW TECO single phase electric motor

^a The data were obtained from the fan nameplate and specifications sheet provided by the requesting party.

^b Based on actual performance test data

Note: ND – No Data, NM – Not Measured

SPECIFICATIONS (Cont'n)

Item		Manufacturer's Specification ^a	AMTEC Verification
5	Heater		
5.1	Oil burner		
5.1.1	Type	Spray-type burner	Spray-type burner
5.1.2	Brand and model	OLYMPIA LT-13	OLYMPIA LT-13
5.1.3	Fuel	Crude oil	Diesel
5.1.4	Heat output, kJ/h	ND	NM
5.1.5	Fuel consumption rate, L/h	3-13 (kg/h)	5.33 ^b
5.1.6	Capacity of fuel tank, L	200	NM
5.1.7	Mechanism for fuel feed	Gear pump	Electronic pump
5.1.8	Method of temperature control	Digital	Digital
5.1.9	Material of construction	ND	NM
5.1.10	Others (specify)	a. Made in Korea b. 150 W, 220 V Electric motor	a. Made in Korea b. 150 W, 220 V Electric motor
5.2	Biomass furnace		
5.2.1	Type	Biomass-fed	Indirect-fired
5.2.2	Brand and model	LAKAS KULIGLIG	Fabricated
5.2.3	Fuel	Biomass (rice hull)	Rice hull
5.2.4	Heat output, kJ/h	ND	419861.08 ^b
5.2.5	Fuel consumption rate, L/h	ND	84.36 (rice hull) ^b
5.2.6	Capacity of fuel tank, L	ND	NM
5.2.7	Mechanism for fuel feed	Plunger-type, automatic feeder	Plunger-bed type
5.2.8	Method of temperature control	Digital	Digital
5.2.9	Material of construction	M.S. plate, Fire tube, refractory ceramic fiber	NM
5.2.10	Others (specify)	a. Equipped with an ash ejector/scrapper to automaticall discharge ash b. Equipped with an ash scrubber to trap fly ash	a. Centrifugal fan is provided for supply of combustion air. b. Furnace is equipped with mechanical system for ash disposal.

^a The data were obtained from the specifications sheet provided by the requesting party.

^b Based on actual performance test data

Note: ND – No Data, NM – Not Measured

SPECIFICATIONS (Cont'n)

Item		Manufacturer's Specification ^a	AMTEC Verification
6	Material handling system		
6.1	Loading hopper/table		
6.1.1	Dimensions, mm		
6.1.1.1	Length	ND	1830
6.1.1.2	Width	ND	1380
6.1.1.3	Height	ND	900
6.1.2	Material of construction	G.I. sheet	Steel sheet
6.1.3	Other features	ND	a. Provided with railings b. Equipped with feeder's shutter to close or open the feeding
6.2	Elevator		
6.2.1	Number	1	1
6.2.2	Height, m	7.50	7.50
6.2.3	Capacity, t/h	12	NM
6.2.4	Belt dimensions, mm	NM	NM
6.2.5	Elevator buckets		
6.2.5.1	Dimensions, mm	190 × 130 × 110	190 × 130 × 110
6.2.5.2	Material of construction	Engineering plastic	Plastic

^a The data were obtained from the specifications sheet provided by the requesting party.

Note: ND – No Data, NM – Not Measured

SPECIFICATIONS (Cont'n)

Item		Manufacturer's Specification ^a	AMTEC Verification
6.2.6	Drive motor		
6.2.6.1	Type	Electric motor	Electric motor
6.2.6.2	Rated power, kW	1.49	NM
6.2.6.3	Rated speed, rpm	1750	NM
6.2.6.4	Phase	Single	Single
6.2.6.5	Current, A	NM	NM
6.2.6.6	Voltage, V	220-230	NM
6.2.6.7	Frequency, Hz	60	NM
6.2.7	Material of construction	ND	ND
6.2.8	Other features	ND	ND
6.3	Unloader		
6.3.1	Type	ND	ND
6.3.2	Capacity, t/h	8	NM
6.3.3	Drive	ND	0.75 kW Electric motor with 1:60 speed reducer
6.3.4	Material	304 stainless steel	Stainless steel
6.4	Bottom conveyor		
6.4.1	Type	ND	Auger
6.4.2	Material	304 stainless steel	Stainless steel
6.4.3	Location	ND	Beneath the unloader
6.5	Grain spreader		
6.5.1	Drive	ND	0.75 kW Electric motor
6.5.2	Location	ND	On top of the drying chamber
7	Instruments and controls		
7.1	Moisture meter	ND	None
7.2	Temperature controller	ND	Digital
7.2.1	Location	ND	On the control panel

^a The data were obtained from the specifications sheet provided by the requesting party.

Note: ND – No Data, NM – Not Measured

SPECIFICATIONS (Cont'n)

Item	Manufacturer's Specification ^a	AMTEC Verification
8 Safety features		
8.1 Product safety	a. With flame sensor. b. With automatic fuel shut-off. c. With adjustable fuel vaporizer. d. With fuel gage. e. With automatic temperature control	Equipped with digital temperature control and sensor for the drying air.
8.2 Machine safety	With emergency stop	Equipped with a main control panel which contains all the push buttons and toggle switches for the different components
9 Operator safety	a. With belt guard b. With protective railing c. With dust collection system	a. Transmission system guard b. Protective railing c. Dust collection system

^a The data were obtained from the specifications sheet provided by the requesting party.

RESULTS

Item		Data	
1	Initial weight of test material, kg	5905.4	
2	Final weight of test material, kg	4859.3	
3	Initial moisture content, % _{wb}	29.2	
4	Final moisture content, % _{wb}	13.9	
5	Actual drying time, h	7	
6	Drying capacity, kg/h	843.63	
7	Rate of moisture removal, kg/h	149.44	
8	Moisture reduction rate, %/h	2.18	
9	Average drying air temperature, °C	71.4	
10	Ambient air condition		
10.1	Average dry bulb temperature, °C	30.8	
10.2	Average wet bulb temperature, °C	25.4	
10.3	Relative humidity, %	65.0	
11	Exhaust air condition		
11.1	Average dry bulb temperature, °C	33.6	
11.2	Average wet bulb temperature, °C	31.2	
11.3	Relative humidity, %	84.0	
12	Average grain temperature, °C	37.3	
13	Fan air velocity, m/s	9.53	
14	Total air flow rate, m ³ /min	166.96	
15	Plenum static pressure, mmH ₂ O	26.05	
16	Heater furnace fuel consumption		
16.1	Biomass furnace, kg/h	84.36	
16.2	Diesel-fed burner, L/h	5.33	
17	Power consumption	Without load	With load
17.1	Total electrical input power, kW	12.40	12.85
17.2	Line voltage, ACV	219.6	219.6
17.3	Load current, A	56.5	58.5
18	Noise level at operator's ear level, dB(A)	Without load	With load
18.1	Feeding operator	73.0	78.2
18.2	Machine operator	71.0	77.0
19	Heat utilization, kJ/kg, water removed	2809.49	
20	Heating system efficiency, %	35.74	
21	Drying efficiency, %	85.43	
22	Drying system efficiency, %	30.53	
23	Moisture content variation at the end of drying, %	0.264	
24	Moisture content gradient, %	1.20	

RESULTS (Cont'n)

Item		Data	
2	Condition of test material	Control ^a	Dryer Test
2.1	Variety	NSIC Rc 402	
2.2	Source	San Rafael, Bulacan	
2.3	Paddy analysis		
2.3.1	Grain size, brown rice, mm		
2.3.1.1	Length	6.71	6.71
2.3.1.2	Width	2.51	2.51
2.3.1.3	Thickness	1.37	1.50
2.3.2	Grain type	Long and bold	Long and bold
2.3.3	Paddy bulk density, kg/m ³	563.24	563.24
2.3.4	Moisture content, % _{wb}	13.7	13.9
2.3.5	Purity, %	97.96	96.34
2.3.6	Cracked, %	0.750	4.50
2.3.7	Immature, %	1.91	1.91
2.8	Chalky, %	0.920	0.920
2.9	Yellow and fermented, %	4.49	4.49
2.10	Red rice, %	0.387	0.387
3	Laboratory milling test		
3.1	Total milling recovery, %	66.67	65.70
3.2	Head rice recovery, %		
3.2.1	Based on milled rice	92.70	90.19
3.2.2	Based on rough rice	61.81	59.26
3.3	Broken rice, %		
3.3.1	Based on milled rice	6.59	8.39
3.3.2	Based on rough rice	4.39	5.51
3.4	Brewer's rice, %		
3.4.1	Based on milled rice	0.708	1.41
		0.472	0.927

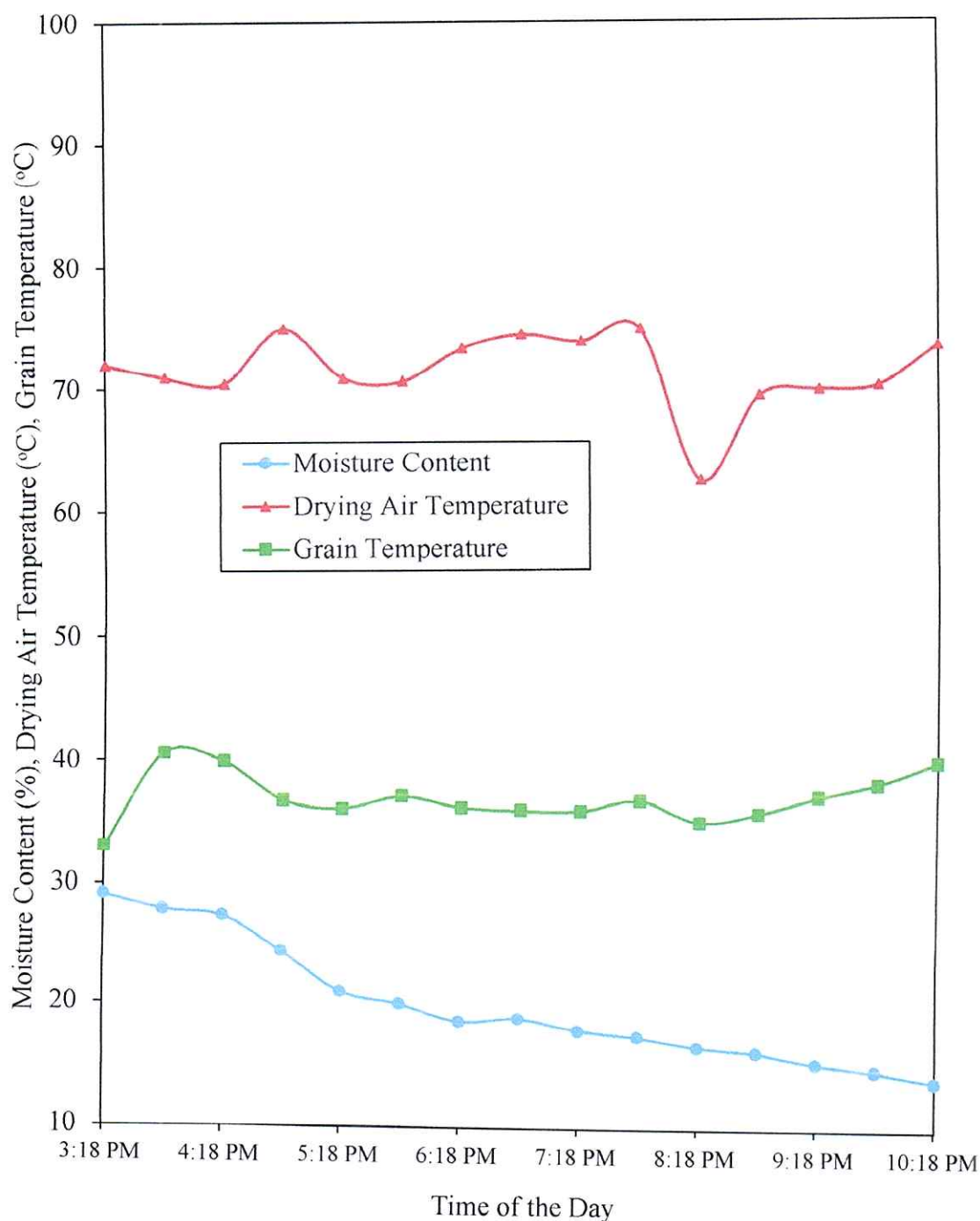
RESULTS (Cont'n)

Figure 3. Drying curve of the LAKAS KULIGLIG RMD-LK66 Heated-Air Mechanical Grain Dryer.

OBSERVATION

Figure 4. Drying chamber still has space after loading only 5905.4 kg of wet paddy.

Tested by:


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Test Engineer


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Test Engineer

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AMTEC Director

28 AUG 2024
Date

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